

Creation Date 06-Apr-2010

Revision Date 14-May-2024

Revision Number 5

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: n-Butyllithium, 2.5M in hexane  
Cat No. : **R16300**  
Molecular Formula C<sub>4</sub> H<sub>9</sub> Li  
REACH registration number -

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

Flammable liquids  
Substances/mixtures which, in contact with water, emit flammable gases  
Pyrophoric liquids

Category 2 (H225)  
Category 1 (H260)  
Category 1 (H250)

## **Health hazards**

Aspiration Toxicity  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Reproductive Toxicity  
Specific target organ toxicity - (single exposure)  
Specific target organ toxicity - (repeated exposure)

Category 1 (H304)  
Category 1 B (H314)  
Category 1 (H318)  
Category 2 (H361f)  
Category 3 (H336)  
Category 2 (H373)

## **Environmental hazards**

Chronic aquatic toxicity

Category 2 (H411)

Full text of Hazard Statements: see section 16

## **2.2. Label elements**



Signal Word

**Danger**

## **Hazard Statements**

H225 - Highly flammable liquid and vapor  
H250 - Catches fire spontaneously if exposed to air  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H304 - May be fatal if swallowed and enters airways  
H314 - Causes severe skin burns and eye damage  
H336 - May cause drowsiness or dizziness  
H361f - Suspected of damaging fertility  
H373 - May cause damage to organs through prolonged or repeated exposure  
H411 - Toxic to aquatic life with long lasting effects  
EUH014 - Reacts violently with water

## **Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P222 - Do not allow contact with air  
P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor/physician

## **2.3. Other hazards**

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

Water reactive

This product does not contain any known or suspected endocrine disruptors

Toxic to terrestrial vertebrates

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component     | CAS No   | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                        |
|---------------|----------|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hexane        | 110-54-3 | EEC No. 203-777-6 | 77       | Flam. Liq. 2 (H225)<br>Asp. Tox. 1 (H304)<br>Skin Irrit. 2 (H315)<br>STOT SE 3 (H336)<br>Repr. 2 (H361f)<br>STOT RE 2 (H373)<br>Aquatic Chronic 2 (H411) |
| Butyl lithium | 109-72-8 | 203-698-7         | 23       | Pyr. Liq. 1 (H250)<br>Water react. 1 (H260)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318)<br>(EUH014)                                                     |

| Component | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-----------|---------------------------------------|----------|-----------------|
| Hexane    | STOT RE 2 (H373) :: C>=5%             | -        | -               |

| REACH registration number |                           | - |
|---------------------------|---------------------------|---|
| Components                | Reach Registration Number |   |
| Butyl lithium             | 01-2119494906-21          |   |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                   |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                   |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                                                                             |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. Risk of serious damage to the lungs (by aspiration). |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                    |

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

## 4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## 4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, fog or alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

Water. Carbon dioxide (CO<sub>2</sub>). Foam.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### **Hazardous Combustion Products**

Butane, Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Handle under an inert atmosphere. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Store under an inert atmosphere. Keep away from water or moist air. Keep refrigerated. Keep away from heat, sparks and flame. Flammables area. Protect from moisture. Keep container tightly closed.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 4.2

**Switzerland - Storage of hazardous substances**

Storage class - SC 4.2  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component | European Union                                       | The United Kingdom                                                                      | France                                                                                                                                                                                                     | Belgium                                                | Spain                                                                          |
|-----------|------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------|
| Hexane    | TWA: 20 ppm (8hr)<br>TWA: 72 mg/m <sup>3</sup> (8hr) | TWA: 72 mg/m <sup>3</sup><br>TWA: 20 ppm<br>STEL: 60 ppm<br>STEL: 216 mg/m <sup>3</sup> | TWA / VME: 20 ppm (8 heures). restrictive limit<br>TWA / VME: 72 mg/m <sup>3</sup> (8 heures). restrictive limit<br>TWA / VME: 1000 mg/m <sup>3</sup> (8 heures).<br>STEL / VLCT: 1500 mg/m <sup>3</sup> . | TWA: 20 ppm 8 uren<br>TWA: 72 mg/m <sup>3</sup> 8 uren | TWA / VLA-ED: 20 ppm (8 horas)<br>TWA / VLA-ED: 72 mg/m <sup>3</sup> (8 horas) |

| Component     | Italy                                                                                                    | Germany                                   | Portugal                                                         | The Netherlands                                                            | Finland                                                               |
|---------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Hexane        | TWA: 20 ppm 8 ore.<br>Time Weighted Average<br>TWA: 72 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average | TWA: 180 mg/m <sup>3</sup><br>TWA: 50 ppm | TWA: 20 ppm 8 horas<br>TWA: 72 mg/m <sup>3</sup> 8 horas<br>Pele | STEL: 144 mg/m <sup>3</sup> 15 minuten<br>TWA: 72 mg/m <sup>3</sup> 8 uren | TWA: 20 ppm 8 tunteina<br>TWA: 72 mg/m <sup>3</sup> 8 tunteina<br>Iho |
| Butyl lithium |                                                                                                          | TWA: 0.2 mg/m <sup>3</sup> (8             |                                                                  |                                                                            |                                                                       |

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

|  |  |                                                                                                                                                                                                                                                                            |  |  |  |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
|  |  | Stunden). MAK<br>inorganic compounds,<br>except Lithium and<br>strong irritant Lithium<br>compounds such as<br>Lithium amide, Lithium<br>hydride, Lithium<br>hydroxide, Lithium<br>nitride, Lithium oxide,<br>Lithium tetrahydro<br>aluminate, Lithium<br>tetrahydroborate |  |  |  |
|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|

| Component | Austria                                                                                                                                                       | Denmark                                                                                                                               | Switzerland                                                                                                                                                   | Poland                                   | Norway                                                                                                                                                                                                                                       |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hexane    | MAK-KZGW: 80 ppm 15<br>Minuten<br>MAK-KZGW: 288 mg/m <sup>3</sup><br>15 Minuten<br>MAK-TMW: 20 ppm 8<br>Stunden<br>MAK-TMW: 72 mg/m <sup>3</sup> 8<br>Stunden | TWA: 20 ppm 8 timer<br>TWA: 72 mg/m <sup>3</sup> 8 timer<br>STEL: 40 ppm 15<br>minutter<br>STEL: 144 mg/m <sup>3</sup> 15<br>minutter | Haut/Peau<br>STEL: 400 ppm 15<br>Minuten<br>STEL: 1440 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 50 ppm 8<br>Stunden<br>TWA: 180 mg/m <sup>3</sup> 8<br>Stunden | TWA: 72 mg/m <sup>3</sup> 8<br>godzinach | TWA: 20 ppm 8 timer<br>TWA: 72 mg/m <sup>3</sup> 8 timer<br>TWA: 40 ppm 8 timer<br>TWA: 275 mg/m <sup>3</sup> 8 timer<br>STEL: 30 ppm 15<br>minutter. value<br>calculated<br>STEL: 108 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated |

| Component | Bulgaria                                   | Croatia                                                                            | Ireland                                                                                                                      | Cyprus                                   | Czech Republic                                                                                                      |
|-----------|--------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Hexane    | TWA: 20 ppm<br>TWA: 72.0 mg/m <sup>3</sup> | kože<br>TWA-GVI: 20 ppm 8<br>satima.<br>TWA-GVI: 72 mg/m <sup>3</sup> 8<br>satima. | TWA: 20 ppm 8 hr.<br>TWA: 72 mg/m <sup>3</sup> 8 hr.<br>STEL: 60 ppm 15 min<br>STEL: 216 mg/m <sup>3</sup> 15<br>min<br>Skin | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 70 mg/m <sup>3</sup> 8<br>hodinách.<br>Potential for cutaneous<br>absorption<br>Ceiling: 200 mg/m <sup>3</sup> |

| Component | Estonia                                                                | Gibraltar                                          | Greece                                   | Hungary                                                                                 | Iceland                                                                                                                               |
|-----------|------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Hexane    | TWA: 20 ppm 8<br>tundides.<br>TWA: 72 mg/m <sup>3</sup> 8<br>tundides. | TWA: 20 ppm 8 hr<br>TWA: 72 mg/m <sup>3</sup> 8 hr | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 72 mg/m <sup>3</sup> 8<br>órában. AK<br>lehetséges borón<br>keresztüli felszívódás | TWA: 20 ppm 8<br>klukkustundum.<br>TWA: 72 mg/m <sup>3</sup> 8<br>klukkustundum.<br>Ceiling: 40 ppm<br>Ceiling: 144 mg/m <sup>3</sup> |

| Component | Latvia                                   | Lithuania                                          | Luxembourg                                                         | Malta                                    | Romania                                              |
|-----------|------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------|------------------------------------------|------------------------------------------------------|
| Hexane    | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 20 ppm IPRD<br>TWA: 72 mg/m <sup>3</sup> IPRD | TWA: 20 ppm 8<br>Stunden<br>TWA: 72 mg/m <sup>3</sup> 8<br>Stunden | TWA: 20 ppm<br>TWA: 72 mg/m <sup>3</sup> | TWA: 20 ppm 8 ore<br>TWA: 72 mg/m <sup>3</sup> 8 ore |

| Component | Russia                                                        | Slovak Republic                                                                          | Slovenia                                                                                                                           | Sweden                                                                                                                                                                | Turkey                                                 |
|-----------|---------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Hexane    | TWA: 300 mg/m <sup>3</sup> 0780<br>MAC: 900 mg/m <sup>3</sup> | Ceiling: 140 mg/m <sup>3</sup><br>TWA: 20 mg/m <sup>3</sup><br>TWA: 72 mg/m <sup>3</sup> | TWA: 20 ppm 8 urah<br>TWA: 72 mg/m <sup>3</sup> 8 urah<br>STEL: 576 mg/m <sup>3</sup> 15<br>minutah<br>STEL: 160 ppm 15<br>minutah | Binding STEL: 50 ppm<br>15 minuter<br>Binding STEL: 180<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 20 ppm 8 timmar.<br>NGV<br>TLV: 72 mg/m <sup>3</sup> 8<br>timmar. NGV | TWA: 20 ppm 8 saat<br>TWA: 72 mg/m <sup>3</sup> 8 saat |

## Biological limit values

List source(s):

| Component | European Union | United Kingdom | France                                                      | Spain                                                 | Germany                                                                                                |
|-----------|----------------|----------------|-------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Hexane    |                |                | 2,5-Hexanedione: 5<br>mg/g creatinine urine<br>end of shift | 2,5-Hexanedione: 0.2<br>mg/L urine end of<br>workweek | 2,5-Hexandione plus<br>4,5-Dihydroxy-2-hexano<br>ne (after hydrolysis): 5<br>mg/L urine (end of shift) |

| Component | Italy | Finland | Denmark | Bulgaria | Romania |
|-----------|-------|---------|---------|----------|---------|
|-----------|-------|---------|---------|----------|---------|

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

|        |  |  |  |  |                                                         |
|--------|--|--|--|--|---------------------------------------------------------|
| Hexane |  |  |  |  | 2,5-Hexandione: 5 mg/g<br>Creatinine urine end of shift |
|--------|--|--|--|--|---------------------------------------------------------|

| Component | Gibraltar | Latvia | Slovak Republic                                                                                                                     | Luxembourg | Turkey |
|-----------|-----------|--------|-------------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| Hexane    |           |        | 2,5-Hexanedione: 5 mg/L urine end of exposure or work shift<br>4,5-Dihydroxy-2-hexanone: 5 mg/L urine end of exposure or work shift |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                 | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|---------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Hexane<br>110-54-3 ( 77 ) |                              |                                 |                                | DNEL = 11mg/kg bw/day             |

| Component                 | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|---------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Hexane<br>110-54-3 ( 77 ) |                                  |                                     |                                    | DNEL = 75mg/m <sup>3</sup>            |

## Predicted No Effect Concentration (PNEC)

No information available.

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Viton (R)      |                                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

## Large scale/emergency use

Any supplied-air respirator that has a full facepiece and is operated in a pressure-demand or other positive pressure mode

## Small scale/Laboratory use

Handle within a fume cupboard or implement suitable equivalent methods to minimize exposure

## Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                                                  |                                          |
|-----------------------------------------|--------------------------------------------------|------------------------------------------|
| Physical State                          | Liquid                                           |                                          |
| Appearance                              | Yellow                                           |                                          |
| Odor                                    | Petroleum distillates                            |                                          |
| Odor Threshold                          | No data available                                |                                          |
| Melting Point/Range                     | No data available                                |                                          |
| Softening Point                         | No data available                                |                                          |
| Boiling Point/Range                     | No information available                         |                                          |
| Flammability (liquid)                   | Highly flammable                                 | On basis of test data                    |
| Flammability (solid,gas)                | Not applicable                                   | Liquid                                   |
| Explosion Limits                        | <b>Lower</b> 1.2 vol %<br><b>Upper</b> 7.8 vol % |                                          |
| Flash Point                             | -21 °C / -5.8 °F                                 | <b>Method -</b> No information available |
| Autoignition Temperature                | No data available                                |                                          |
| Decomposition Temperature               | No data available                                |                                          |
| pH                                      | Not applicable                                   |                                          |
| Viscosity                               | No data available                                |                                          |
| Water Solubility                        | Reacts with water                                |                                          |
| Solubility in other solvents            | No information available                         |                                          |
| Partition Coefficient (n-octanol/water) |                                                  |                                          |
| Component                               | <b>log Pow</b>                                   |                                          |
| Hexane                                  | 4.11                                             |                                          |
| Vapor Pressure                          | 160 mbar @ 20 °C                                 |                                          |
| Density / Specific Gravity              | 0.690                                            |                                          |
| Bulk Density                            | Not applicable                                   | Liquid                                   |
| Vapor Density                           | No data available                                | (Air = 1.0)                              |
| Particle characteristics                | Not applicable (liquid)                          |                                          |

### 9.2. Other information

|                                                                  |                                             |
|------------------------------------------------------------------|---------------------------------------------|
| Molecular Formula                                                | C4 H9 Li                                    |
| Molecular Weight                                                 | 64.06                                       |
| Explosive Properties                                             | Vapors may form explosive mixtures with air |
| Substances/mixtures which, in contact with water, emit flammable | Emitted gas ignites spontaneously           |



# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

gases

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Pyrophoric: Spontaneously flammable in air. Reacts violently with water. Air sensitive. Moisture sensitive.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization Hazardous Reactions

No information available.  
None under normal processing. Reacts violently with water.

### 10.4. Conditions to avoid

Incompatible products. Heat, flames and sparks. Exposure to moist air or water. Exposure to air. Exposure to moisture. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Strong oxidizing agents. Halogens. Bases. Acids. Alcohols.

### 10.6. Hazardous decomposition products

Butane. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

#### Toxicology data for the components

| Component | LD50 Oral              | LD50 Dermal                  | LC50 Inhalation              |
|-----------|------------------------|------------------------------|------------------------------|
| Hexane    | LD50 = 25 g/kg ( Rat ) | LD50 = 3000 mg/kg ( Rabbit ) | LC50 = 48000 ppm ( Rat ) 4 h |

#### (b) skin corrosion/irritation;

Category 1 B

#### (c) serious eye damage/irritation;

Category 1

#### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

#### (e) germ cell mutagenicity;

No data available

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

|                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (f) carcinogenicity;                                                                          | No data available<br>There are no known carcinogenic chemicals in this product                                                                                                                                                                                                                                                                                                   |
| (g) reproductive toxicity;<br>Reproductive Effects<br>Developmental Effects<br>Teratogenicity | Category 2<br>Experiments have shown reproductive toxicity effects on laboratory animals.<br>Developmental effects have occurred in experimental animals.<br>Teratogenic effects have occurred in experimental animals.                                                                                                                                                          |
| (h) STOT-single exposure;<br><br>Results / Target organs                                      | Category 3<br><br>Central nervous system (CNS).                                                                                                                                                                                                                                                                                                                                  |
| (i) STOT-repeated exposure;<br><br>Target Organs                                              | Category 2<br><br>Eyes, Respiratory system, Skin, Gastrointestinal tract (GI), Central nervous system (CNS), Liver, Peripheral Nervous System (PNS).                                                                                                                                                                                                                             |
| (j) aspiration hazard;<br><br>Other Adverse Effects                                           | Category 1<br><br>Tumorigenic effects have been reported in experimental animals.                                                                                                                                                                                                                                                                                                |
| Symptoms / effects, both acute and delayed                                                    | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. |

## 11.2. Information on other hazards

|                                 |                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Endocrine Disrupting Properties | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. Reacts with water so no ecotoxicity data for the substance is available.

| Component | Freshwater Fish                                               | Water Flea          | Freshwater Algae |
|-----------|---------------------------------------------------------------|---------------------|------------------|
| Hexane    | LC50: 2.1 - 2.98 mg/L, 96h flow-through (Pimephales promelas) | EC50: 3.87 mg/L/48h |                  |

### 12.2. Persistence and degradability

#### Persistence Degradability Degradation in sewage treatment plant

Persistence is unlikely, based on information available.  
Reacts with water.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. Water reactive.

### 12.3. Bioaccumulative potential

Product does not bioaccumulate due to reaction with water

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Hexane    | 4.11    | No data available             |

## 12.4. Mobility in soil

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Reacts with water. Will likely be mobile in the environment due to its volatility. Is not likely mobile in the environment. Disperses rapidly in air.

## 12.5. Results of PBT and vPvB assessment

Water reactive.

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors.

## 12.7. Other adverse effects Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance.  
This product does not contain any known or suspected substance.

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

#### Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

#### 14.1. UN number

UN3394

#### 14.2. UN proper shipping name Technical Shipping Name

ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE (N-BUTYLLITHIUM, HEXANE)

#### 14.3. Transport hazard class(es)

4.2

#### Subsidiary Hazard Class

4.3

#### 14.4. Packing group

I

### ADR

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

|                                                                 |                                                                                          |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>14.1. UN number</b>                                          | UN3394                                                                                   |
| <b>14.2. UN proper shipping name</b><br>Technical Shipping Name | ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE<br>(N-BUTYLLITHIUM, HEXANE) |
| <b>14.3. Transport hazard class(es)</b>                         | 4.2                                                                                      |
| Subsidiary Hazard Class                                         | 4.3                                                                                      |
| <b>14.4. Packing group</b>                                      | I                                                                                        |

**IATA** FORBIDDEN FOR IATA TRANSPORT

|                                                                 |                                                                                                                           |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>14.1. UN number</b>                                          | UN3394                                                                                                                    |
| <b>14.2. UN proper shipping name</b><br>Technical Shipping Name | ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE,<br>FORBIDDEN FOR IATA TRANSPORT<br>(N-BUTYLLITHIUM, HEXANE) |
| <b>14.3. Transport hazard class(es)</b>                         | 4.2                                                                                                                       |
| Subsidiary Hazard Class                                         | 4.3                                                                                                                       |
| <b>14.4. Packing group</b>                                      | I                                                                                                                         |

**14.5. Environmental hazards** Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component     | CAS No   | EINECS    | ELINCS    | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|---------------|----------|-----------|-----------|-----|-------|------|----------|------|------|
| Hexane        | 110-54-3 | 203-777-6 | 438-390-3 | -   | X     | X    | KE-18626 | X    | X    |
| Butyl lithium | 109-72-8 | 203-698-7 | -         | -   | X     | X    | KE-04320 | X    | X    |

| Component     | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | AICS | NZIoC | PICCS |
|---------------|----------|------|-----------------------------------------------|-----|-----|------|-------|-------|
| Hexane        | 110-54-3 | X    | ACTIVE                                        | X   | -   | X    | X     | X     |
| Butyl lithium | 109-72-8 | X    | ACTIVE                                        | X   | -   | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component     | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hexane        | 110-54-3 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |
| Butyl lithium | 109-72-8 | -                                                                   | -                                                                             | -                                                                                                     |

REACH links

ALFAAR16300

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component     | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Hexane        | 110-54-3 | Not applicable                                                                            | Not applicable                                                                           |
| Butyl lithium | 109-72-8 | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 2 (self classification)

| Component     | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|---------------|---------------------------------------|-------------------------|
| Hexane        | WGK2                                  |                         |
| Butyl lithium | WGK1                                  |                         |

| Component | France - INRS (Tables of occupational diseases)            |
|-----------|------------------------------------------------------------|
| Hexane    | Tableaux des maladies professionnelles (TMP) - RG 59,RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                 | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|---------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Hexane<br>110-54-3 ( 77 ) | Prohibited and Restricted Substances                                                                           | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H225 - Highly flammable liquid and vapor

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

H250 - Catches fire spontaneously if exposed to air  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H304 - May be fatal if swallowed and enters airways  
H314 - Causes severe skin burns and eye damage  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H336 - May cause drowsiness or dizziness  
H361f - Suspected of damaging fertility  
H373 - May cause damage to organs through prolonged or repeated exposure  
H411 - Toxic to aquatic life with long lasting effects  
EUH014 - Reacts violently with water

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

**Prepared By**

Health, Safety and Environmental Department

**Creation Date**

06-Apr-2010

**Revision Date**

14-May-2024

**Revision Summary**

New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006**

# SAFETY DATA SHEET

n-Butyllithium, 2.5M in hexane

Revision Date 14-May-2024

---

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**